LangGraph Lab Report

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I. Introduction

- What is the role of LangChain?
  LangChain provides tools to connect language models with prompts, data, and external tools.
- What is the role of LangGraph?
  LangGraph organizes an AI workflow using connected steps, shared state, and conditional routing.
- What is the role of an LLM?
  An LLM interprets user requests and generates text to answer questions and follow instructions.

II. Graph Fundamentals

- What is a node?
  A node is a processing step that reads the state and returns updates.
- What is an edge?
  An edge connects nodes and defines the execution flow.
- What is START?
  START is the entry point of the graph.
- What is END?
  END marks the end of graph execution.
- Why is the state passed between nodes?
  The state shares information and intermediate results between processing steps.

III. Routing and Tool Use

The introductory router directs questions to mathematics, programming, or artificial intelligence agents. The tool-enabled workflow uses calculator and current datetime. Execution traces show the sequence: tool request, tool result, and final answer.

IV. Memory

Add_messages accumulates messages, while MemorySaver preserves conversation state across calls with the same thread_id. The assistant remembered “Ahmed” in the same thread but not in a different one.

V. Visualizing and Testing Graphs

Graph visualizations show the nodes, routing paths, and tool-execution loop. The final assistant’s diagram was exported for submission.

VI. Evaluation and Discussion

The introductory router correctly classified all six questions, achieving 100% routing accuracy. No failures occurred in this test set, but six examples cannot demonstrate general reliability or answer correctness.

Structured output, enum validation, few-shot examples, and more ambiguous test questions would improve routing robustness.

Discussion: Where did the router fail, and how could it improve?

The router made no errors on the six test questions, achieving 100% routing accuracy. However, this small test set does not establish reliability on unseen questions or measure the correctness of generated answers.

Routing could be improved through structured output, few-shot examples, and enum validation to restrict responses to valid categories. Additional tests should include ambiguous and multi-domain questions.

VII. Intelligent University Assistant

1. Classifier

Questions are classified into COURSES, PROGRAMMING, MATHEMATICS, ADMINISTRATION, or GENERAL. Empty inputs and classifier failures trigger fallback behavior.

2. Specialized Agents

Each category has a dedicated system prompt. Agents answer questions and handle model-call exceptions.

3. Routing and Graph Assembly

Conditional edges select the appropriate agent. Tool requests pass through ToolNode, then return to the selected agent. MemorySaver supports conversational continuity.

4. Demonstration

Tests cover all five categories, empty input, memory, and tools. The calculator returned 126 for (15 + 27) * 3, and the assistant recalled the user's name.

5. Optional Extension

Wikipedia was integrated using @tool. An initial JSON decoding error was followed by successful tests after configuring a custom User-Agent. The final trace confirms retrieval and a generated summary.

Conclusion

The notebook demonstrates routing, tools, memory, and error handling.